

# **Project Proposal: Mining EV Registration Patterns in Washington State**

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## **Problem**

Electric vehicles (EVs) are becoming an increasingly important part of the transportation system, yet the registered EV population is not distributed uniformly across vehicle types, manufacturers, model years, electric ranges, and geographic areas. As humans are becoming more concerned about the environment and sustainability, the demand and use of Electric Vehicles is increasing every day. Their use in the USA is also increasing accordingly, making such vehicles an important part of transportation here. The population of Electric Vehicles (EV's) is not uniformly distributed across locations, and they differ in vehicle types, models, make, year of production, electric range, etc.

For this project, we have decided to explore electric vehicle use in the state of Washington and identify some meaningful trends and patterns in the data. We can explore the relationships and the differences among vehicles regarding their make, model, electric range, and other characteristics. Understanding these differences can provide useful insight into how EV populations vary across regions and what vehicle characteristics are common, distinctive, or unusual. The output knowledge may support better transportation planning, charging-infrastructure analysis, a future studies of electric vehicle populations.

## **Selected Domain**

The project falls within the transportation and public/government data domain, with a specific focus on electric vehicle population characteristics and geographic variation in Washington State.

- Domain: Transportation
- Source: Washington State Department of Licensing, "Electric Vehicle Population Data" (data.wa.gov)
- Size: ~290,000 records; we will use a fixed random sample of 50,000 rows (seed 42) for manageability.
- Attributes (~17): county, city, ZIP code, model year, make, model, EV type, electric range, base MSRP, legislative district, electric utility, vehicle location, among others

## **Dataset and Source**

The project will use the Electric Vehicle Population Dataset from the Washington State Department of Licensing, available through the Washington State open-data portal. The dataset contains around 300k records and 17 attributes such as county, city, ZIP code, model year, make, model, electric vehicle type, electric range, CAFV eligibility, legislative district, vehicle location, and electric utility. It includes both Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs).

A preliminary inspection also identified data-quality and preprocessing issues that can be investigated later in the project. These include missing values in several attributes, a small number of records located outside Washington, and many zero values for Electric Range that appear to represent unavailable or unresearched range information rather than actual zero-mile vehicle ranges. As this dataset consists of many records and multiple categorical, numerical, and geographical attributes, and it requires genuine preprocessing decisions, it is quite appropriate for data mining.

## **Unit of Analysis**

The primary unit of analysis is an individual registered electric vehicle. Each row represents one vehicle and contains information about the characteristics of the vehicle as well as its registered location. We may aggregate individual vehicles by county or ZIP code to create geographic features such as the number of registered EV's, percentage of BEV's and PHEV's, average model year, and average electric range.

- Vehicle-level for anomaly detection and association rule mining
- ZIP-code level (aggregated fleet features) for clustering

## **Initial Mining Questions**

- What distinct fleet profiles emerge when ZIP codes are grouped by vehicle type, range, and model year?
- Which combinations of vehicle attributes occur more frequently than expected by chance?
- Which vehicle records show anomalous range or model-year values, and do they reflect genuine outliers or data errors?
- How does the relationship between model year and range differ between BEVs and PHEVs?